

For instance, instead of writing

$$x^2 + y^2 = -1 \text{ has no real solutions.}$$

you can write

$$\text{The equation } x^2 + y^2 = -1 \text{ has no real solutions.}$$

It is conventional in mathematics to write in first-person plural, using “we” and “us.” In mathematics, “we” is not used royally; rather, it stands for “the reader and I.” The reader is part of the journey too. Using the first-person plural helps us avoid writing in the passive voice (e.g. “it is shown that”), which is frowned upon in mathematics.

1.7 Problems

(1) Which of the following statements are equivalent to which others?

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|-----------------------|-------------------------------|
| (a) If P then Q . | (f) Whenever P , Q . |
| (b) If Q then P . | (g) P implies Q . |
| (c) P only if Q . | (h) Not P implies not Q . |
| (d) Q only if P . | (i) Not Q implies not P . |
| (e) P if Q . | |

(2) Give an example of statements P and Q , such that P implies Q , but Q does not imply P . Given a statement “ P implies Q ,” we call the statement “ Q implies P ” the *converse*. Statements and their converses are not the same!

(3) Given a statement “if P then Q ,” the statement “if not P then not Q ” is called the *inverse*. Explain why the inverse and converse are always equivalent: if one of them is true, then so is the other; and if one of them is false, then so is the other.

(4) Suppose that $P(x, y)$ is a statement about two numbers x and y . Are the following necessarily equivalent: $\forall x : \exists y : P(x, y)$ and $\exists y : \forall x : P(x, y)$? Either prove they are equivalent for all statements P , or else give an example of a statement P such that one of them is true and the other is false. (Make sure you understand why this problem has content! You cannot conclude that they are equivalent just because they contain the same symbols, albeit in a different order!)

- (5) Prove the following statements:
- (a) For all integers x , $x^2 + 5x$ is even.
 - (b) If x is odd, then $x^3 - 3x^2 + 5x + 7$ is even.
 - (c) Suppose x is an integer. If $x^2 + 2x + 3$ is even, then x is odd.
 - (d) For all integers n , either n^2 or $n^2 - 1$ is a multiple of 4.
- (6) Find, with proof, the largest integer n such that, for all integers x , $x(x+1)(x+2)(x+3)$ is a multiple of n .
- (7) Mimic the proof of the irrationality of $\sqrt{2}$ to prove that $\sqrt{3}$ and $\sqrt[3]{4}$ are irrational. What goes wrong when you try to use the same argument to prove that $\sqrt{9}$ is irrational?
- (8) Prove that the sum of a rational number and an irrational number is irrational. Prove that there exist two irrational numbers whose sum is rational.
- (9) Prove that the product of a nonzero rational number and an irrational number is irrational. Prove that there exist two irrational numbers whose product is rational.
- (10) Prove that $\log_2(3)$ is irrational. (In case you haven't seen logarithms yet, $\log_2(3)$ is the number x such that $2^x = 3$.)
- (11) Note that $(\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = 2$. Explain how to use this fact to prove that there exist two irrational numbers x and y such that x^y is rational.
- (12) Prove that there are no integers x and y such that $x^2 - 3y^2 = 2$.

*P and Q are both arbitrary labels for statements.
 9/22-27/26
 Alexander

Transition to Proofs week 1

1.

P	Q	If P then Q	If Q then P	P only if Q
T	T	T	T	T
F	F	T	T	T
T	F	F	T	F
F	T	T	F	T

Q only if P	P if Q	Whenever P, Q	P implies Q	Not P implies not Q
T	T	T	T	T
T	T	T	T	T
T	F	F	F	F
F	T	T	T	T

Not Q implies not P
T
T
T
F

Statements labeled
 A in the truth table
 are all equivalent, and
 vice versa for B statements.

2. A classic with squares: $x=1$ implies that $x^2=1$, but $x^2=1$ does not imply that $x=1$ (x can also be -1).

3. Let us recall that contrapositives of statements are equivalent.

If P then Q = If not Q then not P,
 for example. But this works when swapping
 P and Q: If Q then P = If not P then not Q.
 These happen to be the converse and inverse
 of If P then Q, so the converse = the inverse.

* This is mostly arbitrary; a few statements are bad to use, but most are fine

4. Proof by Contradiction. Say that $P(x,y)$ is that $x+y=10$. Assume that $\forall x: \exists y: x+y=10$ is equivalent to $\exists y: \forall x: x+y=10$. We need to show that both hold true.

First statement: $y=10-x$. For any x chosen, a y will result. True ☒.

Second statement: $y=10-x$. When we "choose" a y , we will be able to find a corresponding x , but changing the value of that x by any amount makes the solution false - but statement 2 would still be true, which is a contradiction. Therefore, statement 2 is False ☒, so the two statements aren't equal. (In other words, order matters!)

5. a) Assume the contrary, where there exists an x^2+5x that is odd for a specific x .
 $x^2+5x=2m+1$. If x is even ($x=2k$):
 $4k^2+10k=2m+1$. Factor by 2 both sides:
 $2(2k^2+5k)=2(m+0.5)$. Uh oh, $m+0.5$ can't be an integer if m is, so x cannot be even.
If x is odd, $(4k^2+4k+1)+10k+1=4k^2+14k+2$, which is again even while $2m+1$ is not.
 x can't be odd. Therefore, x^2+5x must always be even.

b) Say that $x = 2k + 1$ (odd.) Then $x^3 - 3x^2 + 5x + 7 = \dots$

$$\dots (4k^2 + 4k + 1)(2k + 1) - 12k^2 - 12k - 3 + 10k + 5 + 7$$

$$= 8k^3 + 12k^2 + 6k + 1 - 12k^2 - 2k + 9$$

$$= 8k^3 + 4k + 10 = 2(4k^3 + 2k + 5), \text{ therefore the polynomial is even when } x \text{ is odd.}$$

c) For $x^2 + 2x + 3$ to be even, x^2 must be odd because odd + even + odd = even.

If x^2 is odd, then x is odd because squaring an odd number gives an odd number.

d) If n is even, it already has a factor of 2, which becomes $2^2 = 4$ when squared, so n^2 is div. 4.

If n is odd, then it is either $4k - 1$ or $4k + 1$. Squaring these:

$$(4k - 1)^2 = 16k^2 - 8k + 1 = 4(4k^2 - 2k) + 1.$$

$$(4k + 1)^2 = 16k^2 + 8k + 1 = 4(4k^2 + 2k) + 1.$$

Both of these are in the form $4k + 1$, so $n^2 - 1$ is a multiple of 4 if n is odd.

6. The product of four consecutive integers must:

- have two even numbers, one a multiple of 4, and thus a factor of $2 \cdot 4 = 8$
- have at least one number divisible by 3

Multiplying these factors gives 24, which happens to be $4!$. Any number factorable as $x(x+1)(x+2)(x+3)$ must be div. by 24.

7. Proof by Contradiction. Assume $\sqrt{3} = \frac{a}{b}$, where a and b are integers as small as possible. Thus $3 = \frac{a^2}{b^2}$ and $3b^2 = a^2$.

This implies that a^2 is divisible by 3, so $a^2 = (3c)^2 = 9c^2 = 3b^2$. $b^2 = 3c^2$, $3 = \frac{b^2}{c^2}$, where $0 < c < b$. But we found a rational number less on each side than $\frac{a}{b}$, $\frac{b}{c}$. This is a contradiction, so $\sqrt{3}$ is irrational.

Corollary. Assume $\sqrt[3]{4} = \frac{a}{b}$, $4 = \frac{a^3}{b^3}$, $4b^3 = a^3$. a^3 must be a multiple of 4, so an even factor of a exists. say $a = 4c$, so $4b^3 = (4c)^3 = 16c^3$, and $b^3 = 4c^3$, so $4 = \frac{b^3}{c^3}$, $\sqrt[3]{4} = \frac{b}{c}$.

Again, we find the contradiction, and $\sqrt[3]{4}$ is irrational. Now to try $\sqrt{9}$. Assume $\sqrt{9} = \frac{a}{b}$ (it does)...

$$9 = \frac{a^2}{b^2}, 9b^2 = a^2 = (3c)^2 = 9c^2, b^2 = c^2, 1 = \frac{b^2}{c^2}.$$

Uh... couldn't find that a more simplified ratio exists (just became 1), so $\sqrt{9} = 3$ is rational.

8. Assume the contrary: $\frac{a}{b} + n = \frac{c}{d}$, where n is irrational. $n = \frac{c}{d} - \frac{a}{b} = \frac{cb - ad}{bd}$. However, since a, b, c , and d are integers, $\frac{cb - ad}{bd}$ is rational, a contradiction since n must be irrational. Therefore, $\frac{a}{b} + n = m$ (rational). As for the second sentence, try $\pm \pi$ and $-\pi$, or $\pm \sqrt{2}$.

9. Contrary: $\frac{a}{b} \cdot n = \frac{c}{d}$, $n = \frac{c}{d} \div \frac{a}{b} = \frac{c}{d} \cdot \frac{b}{a} = \frac{cb}{ad}$. But $\frac{cb}{ad}$ is rational and n isn't, so a contradiction forms. As for sentence 2, try π and $\frac{1}{\pi}$, or $\sqrt{2}$ and $\frac{1}{\sqrt{2}}$.

10. Say that $\log_2(3) = \frac{a}{b}$. Thus, $2^{\frac{a}{b}} = 3$ and $2^a = 3^b$.
 The only number that satisfies this is $1=1$ (1 is both a power of 2 and 3), or $2^0 = 3^0$. But $\frac{0}{0}$ is n't rational - it's undefined. Therefore, $\log_2(3)$ is NOT rational.

11. Case 1: $\sqrt{2}^{\sqrt{2}}$ is rational

We're done! $\sqrt{2}^{\sqrt{2}}$ is a great example.

Case 2: $(\sqrt{2}^{\sqrt{2}})^{\sqrt{2}}$ is irrational

$(\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = 2$, which is an irrational raised to an irrational. Example found again!

(Advanced math shows that only case 2 is true, but that's beyond the scope of T+P.)

12. Case 1: $x = 3k$

$$9k^2 - 3y^2 = 2 = 3(3k^2 - y^2). \quad 3k^2 - y^2 = \left(\frac{2}{3}\right), \text{ not an integer!}$$

Case 2: $x = 3k + 1$

$$9k^2 + 6k + 1 - 3y^2 = 2, \quad 3(3k^2 + 2k - y^2) = 1, \quad 3k^2 + 2k - y^2 = \left(\frac{1}{3}\right) \text{ contradiction!}$$

Case 3: $x = 3k + 2$

$$9k^2 + 12k + 4 - 3y^2 = 2, \quad 3(3k^2 + 4k - y^2) = -2, \quad 3k^2 + 4k - y^2 = \left(-\frac{2}{3}\right), \text{ not an integer!}$$

All three cases of $x \bmod 3$ show that 2 never shows up from integer x 's and y 's. Thus, $x^2 - 3y^2 = 2$ is false $\forall (x, y)$.

~ Q. E. D. ~